

Gabriel Deza

PhD Student, Industrial Engineering, Tel Aviv University

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Summary

PhD student in Industrial Engineering at Tel Aviv University working with Tal Raviv and Michal Tzur. Previously completed an MSc in Industrial Engineering and Operations Research at the University of California, Berkeley with Candace Yano and a BSc in Engineering Science at the University of Toronto, including research with Nicolas Papernot and the CleverHans Lab. Current research focuses on optimization problems for transportation systems, especially line planning, timetabling, and demand routing in highly connected networks, as well as exact optimization methods spanning multiple stages of the network design problem.

Education

PhD in Industrial Engineering

Tel Aviv University

Jan 2023 – Present

Tel Aviv, Israel

- Advised by Tal Raviv and Michal Tzur.
- Research focus: transportation optimization, line planning, timetabling, and demand routing in highly connected networks.

MSc in Industrial Engineering and Operations Research

University of California, Berkeley

Aug 2021 – Aug 2022

Berkeley, CA

BASc in Engineering Science, Major in Machine Learning

University of Toronto

Sept 2017 – May 2021

Toronto, Canada

Research Experience

Researcher

Industrial Engineering and Operations Research, UC Berkeley

May 2022 – Sept 2022

Berkeley, CA

- Worked with Candace Yano on food delivery pricing models and marketplace design.
- Developed revenue management models for customer, restaurant, and driver interactions to study profit-maximizing service fees and payments.

Researcher

AutoLab with Google Robotics, UC Berkeley

May 2021 – May 2022

Berkeley, CA

- Worked with Ken Goldberg and Google Robotics on autonomous garment folding.
- Trained and evaluated policies in SoftGym for simulated and physical cloth-manipulation tasks.

Researcher

CleverHans Lab, University of Toronto

May 2020 – May 2021

Toronto, Canada

- Worked with Nicolas Papernot on machine unlearning and the robustness of financial trading systems.
- Developed machine learning formulations that facilitate machine unlearning requests effectively and tractably.

Researcher

Technion – Israel Institute of Technology

May 2019 – Aug 2019

Haifa, Israel

- Worked with Shmuel Onn on graph optimization algorithms.
- Developed dynamic programming formulations for optimization over subgraph degree objectives.

Researcher

Université Paris-Saclay

May 2018 – Aug 2018

Paris, France

- Worked with Yannis Manoussakis on topological graph algorithms.
- Implemented graph algorithms and LP/IP models in Python, Gurobi, and CPLEX.

Publications

Routing in Line Networks with Handling Times. Gabriel Deza, Tal Raviv, Michal Tzur. Submitted to *Transportation Science*, 2026.

Optimizing Service Fees and Driver Payments for Restaurant Food Delivery. Gabriel Deza, Candace Yano. Under review at *Manufacturing & Service Operations Management (MSOM)*, 2025.

Learning to Fold Real Garments with One Arm: A Case Study in Cloud-Based Robotics Research. Ryan Hoque, Kaushik Shivakumar, Shrey Aeron, **Gabriel Deza**, Aditya Ganapathi, Adrian Wong, Johnny Lee, Andy Zeng, Vincent Vanhoucke, Ken Goldberg. Accepted to *IROS*, 2022.

Unrolling SGD: Understanding Factors Influencing Machine Unlearning. Anvith Thudi*, **Gabriel Deza***, Varun Chandrasekaran, Nicolas Papernot. Accepted to *EURO S&P*, 2022.

Interpretability in Safety-Critical Financial Trading Systems. **Gabriel Deza***, Adelin Travers, Colin Rowat, Nicolas Papernot. *arXiv* preprint, 2021.

Optimization over Degree Sequences of Graphs. **Gabriel Deza**, Shmuel Onn. *Discrete Applied Mathematics* 296 (2021), 2–8.

* denotes primary or co-primary authorship.

Teaching

- Fall 2023: 0512-1820 C Programming
- Winter 2024: 0571-4133 Geographic Network Systems
- Fall 2025: 0571-4182 Introduction to Deep Learning
- Winter 2026: 0571-4184 Advanced Topics in Deep Learning

Skills

- Programming: Python, C, Matlab, Julia, SQL
- ML/OR tools: PyTorch, TensorFlow, Gurobi, CPLEX

Awards

- ISTRC Conference Grant for TSL Workshop 2024
- ISTRC Scholarship, 2023–2024
- Shlomo Smeltzer Scholarship, 2023 and 2024
- UC Berkeley IEOR Fellowship, 2021–2022
- UC Berkeley Fellowship Nominee, 2021
- University of Toronto ECE Studentship Award, 2020
- Exceptional Opportunity Award, 2019
- Exceptional Opportunity Award, 2018
- First Place, University of Toronto UTRA Hackathon, 2018
- Faculty of Applied Science and Engineering Admission Scholarship, 2017

Languages

- English and French (bilingual)
- Hebrew (beginner)